Business Performance Analysis

Kimia Farma - Big Data Analytics

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Disclaimer

This final project was completed as part of a project-based Business Intelligence Analyst Virtual Internship program, utilizing a structured dataset and an analytical approach supported by data visualization.

About Company

Kimia Farma is one of Indonesia's largest integrated pharmaceutical company, operating across the pharmaceutical value chain, including drug manufacturing, distribution, retail pharmacy services, clinical services, and healthcare products. With an extensive network of branches across the country, the company processes thousands of transactions every day, generating large volumes of business data.

Project Background

In today's competitive pharmaceutical industry, business performance extends beyond revenue growth and is also measured by transaction volume, profitability, customer satisfaction, and regional performance. Leveraging business data enables companies to evaluate operational performance, monitor business growth, identify improvement opportunities, and support data-driven decision-making.

The dataset available are **kf_final_transaction** (transaction records), **kf_product** (product information), **kf_kantor_cabang** (branch information), and **kf_inventory** (inventory data). These datasets provide comprehensive information on sales, products, customer transactions, branch operations, and inventory management.

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Problem Statement

Kimia Farma operates an extensive pharmacy network across Indonesia, generating large volumes of transactional data through its daily operations. To support business growth and operational excellence, management requires a comprehensive understanding of business performance across different regions and time periods.

However, without systematic analysis, it is difficult to evaluate revenue growth trends, identify the provinces contributing the highest transaction volume and net sales, detect branches with discrepancies between branch performance and customer transaction experience, and understand the geographical distribution of profitability across Indonesia.

This project addresses the following business questions:

- How is PT Kimia Farma revenue trend from 2020 to 2023?
- Which are the provinces with the highest total transaction volume?
- Which are the provinces generating the highest net sales?
- Which are the branches with the highest branch ratings but the lowest transaction ratings?
- How is Net Profit distributed across provinces in Indonesia?

By answering these business questions, the analysis aims to provide actionable insights that support data-driven decision-making, improve operational performance, and drive sustainable business growth.

Goals

Objectives

- Analyzing **year-over-year revenue trends** from 2020-2023.
- Identifying the **Top 10 provinces** by transaction volume.
- Identifying the **Top 5 provinces** by Net Sales.
- Evaluating branches with **high branch ratings but low transaction ratings**.
- Analyzing the **geographical distribution of Net Profit** across Indonesia.
- Developing an **interactive dashboard** to support business monitoring and strategic decision-making.

Importing Dataset to BigQuery

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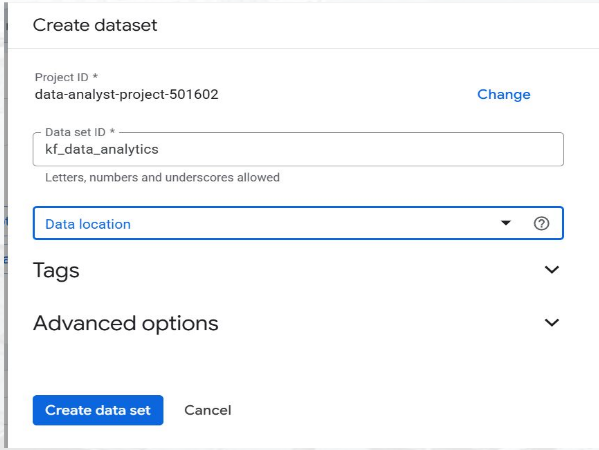
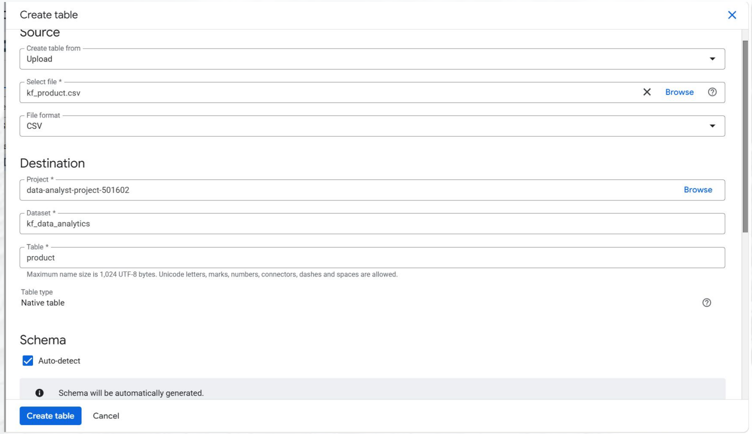
The first step was importing the four CSV datasets (**kf_final_transaction**, **kf_product**, **kf_kantor_cabang**, and **kf_inventory**) into **Google BigQuery**. A dataset named **kf_data_analytics** was created, and each CSV file was uploaded as a separate table using the **Create Table** feature with **Schema Auto Detect** enabled.





Successfully imported dataset to BigQuery:

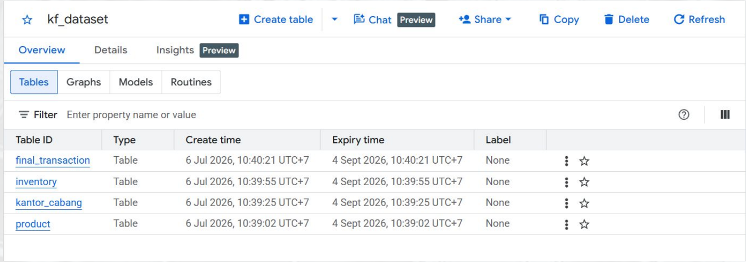


Table ID	Type	Create time	Expiry time	Label
final_transaction	Table	6 Jul 2026, 10:40:21 UTC+7	4 Sept 2026, 10:40:21 UTC+7	None
inventory	Table	6 Jul 2026, 10:39:55 UTC+7	4 Sept 2026, 10:39:55 UTC+7	None
kantor_cabang	Table	6 Jul 2026, 10:39:25 UTC+7	4 Sept 2026, 10:39:25 UTC+7	None
product	Table	6 Jul 2026, 10:39:02 UTC+7	4 Sept 2026, 10:39:02 UTC+7	None

Once imported, the data was ready for SQL queries, table joins, and business performance analysis.

Entity Relationship Diagram

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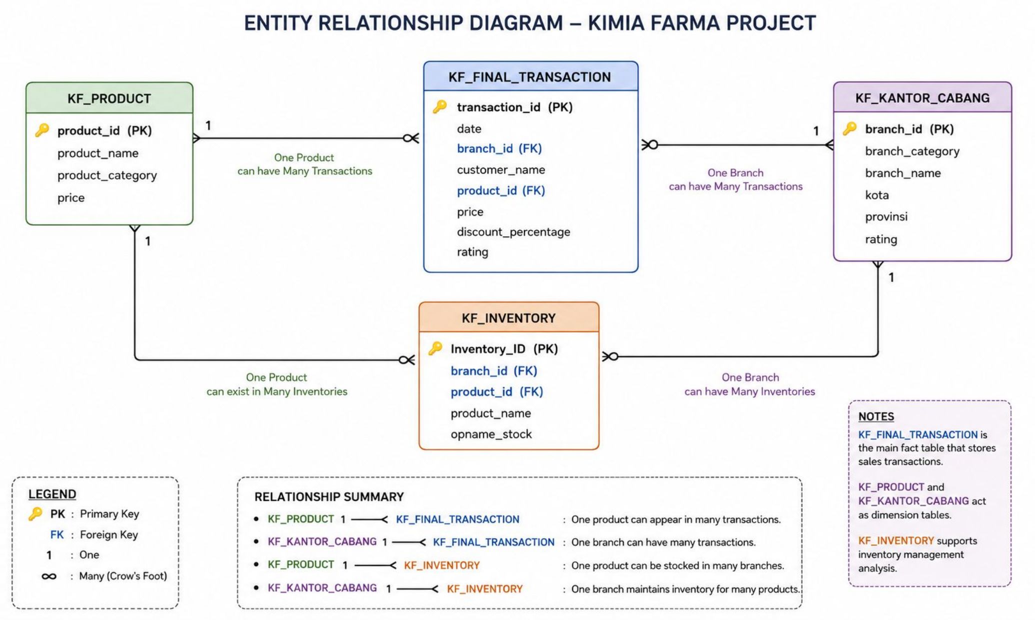

kf_product → **kf_final_transaction**: One-to-Many (1:M)

kf_kantor_cabang → **kf_final_transaction**: One-to-Many (1:M)

kf_product → **kf_inventory**: One-to-Many (1:M)

kf_kantor_cabang → **kf_inventory**: One-to-Many (1:M)

Analysis Table

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In this project, an **analysis table** is created by aggregating and integrating data from the four previously imported datasets. The resulting table serves as the primary dataset for business performance analysis and include the following columns:

transaction_id : Unique identifier for each transaction.

date : Date when the transaction was made.

branch_id : Unique identifier for the Kimia Farma branch.

branch_name : Name of the Kimia Farma branch.

kota : City where the branch is located.

provinsi : Province where the branch is located.

rating_cabang : Customer rating of the Kimia Farma branch.

customer_name : Name of the customer who made the transaction.

product_id : Unique identifier for the product.

product_name : Name of the product.

actual_price : Original product price before discount.

discount_percentage : Discount percentage applied to the product.

gross_profit_percentage : Gross profit percentage determined based on the product price:

- Price ≤ IDR 50,000 → **10%**
- IDR 50,000 < Price ≤ IDR 100,000 → **15%**
- IDR 100,000 < Price ≤ IDR 300,000 → **20%**
- IDR 300,000 < Price ≤ IDR 500,000 → **25%**
- Price > IDR 500,000 → **30%**

net_sales : Sales value after applying the discount.

net_profit : Profit earned by Kimia Farma, calculated using the corresponding gross profit percentage.

rating_transaksi : Customer rating of the transaction.

BigQuery Syntax

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```
SELECT t.transaction_id, t.date, t.branch_id, kc.branch_name, kc.kota, kc.provinsi, kc.rating AS rating_cabang, t.customer_name,
t.product_id, p.product_name, t.price AS actual_price, t.discount_percentage,
    CASE
        WHEN t.price <= 50000 THEN 0.10
        WHEN t.price <= 100000 THEN 0.15
        WHEN t.price <= 300000 THEN 0.20
        WHEN t.price <= 500000 THEN 0.25
        ELSE 0.30
    END AS gross_profit_percentage,
    t.price * (1 - t.discount_percentage) AS nett_sales,
    (t.price * (1 - t.discount_percentage)) *
    CASE
        WHEN t.price <= 50000 THEN 0.10
        WHEN t.price <= 100000 THEN 0.15
        WHEN t.price <= 300000 THEN 0.20
        WHEN t.price <= 500000 THEN 0.25
        ELSE 0.30
    END AS nett_profit,
    t.rating AS rating_transaksi
FROM `data-analyst-project-501602.kf_dataset.final_transaction` t
JOIN `data-analyst-project-501602.kf_dataset.kantor_cabang` kc
    ON t.branch_id = kc.branch_id
JOIN `data-analyst-project-501602.kf_dataset.product` p
    ON t.product_id = p.product_id;
```

Analysis Table Result

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Query results | Chat | View all jobs | Save results | Open in

Job information | Results | Visualisation | JSON | Execution details | Execution graph

transaction_id	date	branch_id	branch_name	kota	provinsi
TRX5103706	2021-08-25	93529	Kimia Farma - Klinik & Apotek	Yogyakarta	DI Yogyakarta
TRX5388139	2020-12-29	24832	Kimia Farma - Klinik-Apotek-Lab...	Pekanbaru	Riau
TRX7251897	2020-02-03	20505	Kimia Farma - Apotek	Cilacap	Jawa Tengah
TRX4943675	2022-09-09	17678	Kimia Farma - Klinik & Apotek	Subang	Jawa Barat
TRX3469820	2020-06-20	28315	Kimia Farma - Klinik-Apotek-Lab...	Sukabumi	Jawa Barat
TRX1213133	2021-09-17	22280	Kimia Farma - Apotek	Batam	Kepulauan Riau
TRX2020131	2020-12-16	40028	Kimia Farma - Klinik-Apotek-Lab...	Balikpapan	Kalimantan Timur
TRX5015870	2022-08-17	41343	Kimia Farma - Apotek	Semarang	Jawa Tengah
TRX7064077	2021-06-21	86546	Kimia Farma - Klinik & Apotek	Pematangsiantar	Sumatera Utara
TRX5979742	2020-12-31	18235	Kimia Farma - Klinik & Apotek	Pekanbaru	Riau
TRX2209141	2021-03-20	59571	Kimia Farma - Klinik-Apotek-Lab...	Tarakan	Kalimantan Utara
TRX5385534	2023-03-17	69280	Kimia Farma - Klinik-Apotek-Lab...	Purwakarta	Jawa Barat

Query results | Chat | View all jobs | Save results

on | Results | Visualisation | JSON | Execution details

gross_profit_percentage	nett_sales	nett_profit	rating_transaksi
0.2	249183.0	49836.60000000...	3.0
0.2	226530.0	45306.0	3.0
0.2	213945.0	42789.0	3.0
0.2	213945.0	42789.0	3.0
0.2	226530.0	45306.0	3.0
0.2	216462.0	43292.4	3.0
0.2	241632.0	48326.4	3.0

Query results | Chat | View all jobs | Save results | Open in

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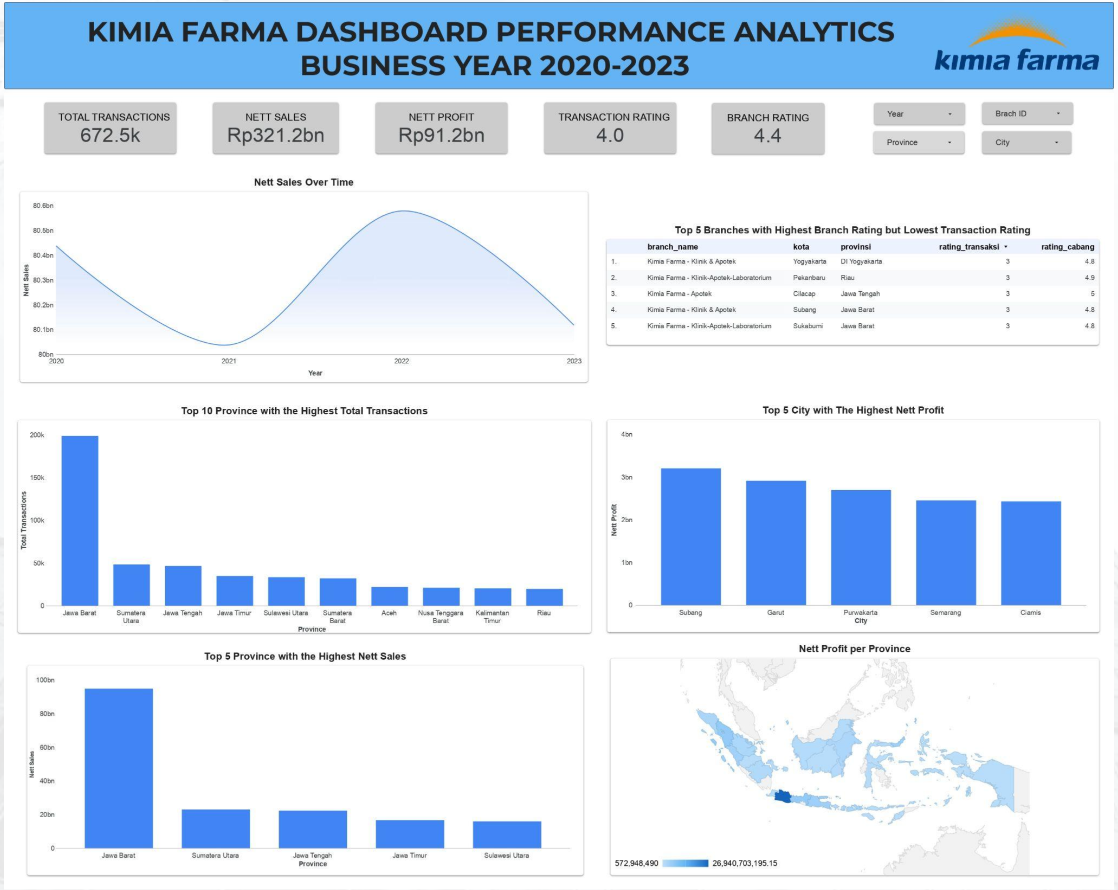
rating_cabang	customer_name	product_id	product_name	actual_price	discount_percent...
4.3	Derrick Wright III	KF116	Psycholeptics drugs, Hypnotics ...	251700	0.1
4.2	Elizabeth Ramos	KF116	Psycholeptics drugs, Hypnotics ...	251700	0.12
4.5	Meghan Warner	KF116	Psycholeptics drugs, Hypnotics ...	251700	0.09
4.8	Steven Roberts	KF116	Psycholeptics drugs, Hypnotics ...	251700	0.1
3.9	Linda Bruce DDS	KF116	Psycholeptics drugs, Hypnotics ...	251700	0.07
4.5	Cory Castro	KF116	Psycholeptics drugs, Hypnotics ...	251700	0.11
4.0	Stephanie Boone	KF116	Psycholeptics drugs, Hypnotics ...	251700	0.03
4.0	Mary Hughes	KF116	Psycholeptics drugs, Hypnotics ...	251700	0.03
4.5	Tamara Bruce	KF116	Psycholeptics drugs, Hypnotics ...	251700	0.04
4.8	Aaron Reed	KF116	Psycholeptics drugs, Hypnotics ...	251700	0.11
4.9	Nancy Kennedy	KF116	Psycholeptics drugs, Hypnotics ...	251700	0.1
4.0	Paul Morales	KF116	Psycholeptics drugs, Hypnotics ...	251700	0.11

Kimia Farma Dashboard Performance Analytics



	branch_name	kota	provinsi	rating_transaksi	rating_cabang
1.	Kimia Farma - Klinik & Apotek	Yogyakarta	DI Yogyakarta	3	4.8
2.	Kimia Farma - Klinik-Apotek-Laboratorium	Pekanbaru	Riau	3	4.9
3.	Kimia Farma - Apotek	Cilacap	Jawa Tengah	3	5
4.	Kimia Farma - Klinik & Apotek	Subang	Jawa Barat	3	4.8
5.	Kimia Farma - Klinik-Apotek-Laboratorium	Sukabumi	Jawa Barat	3	4.8

Analysis & Insights

Nett Sales Trend (2020-2023)

Insight:

- **Net sales declined from 2020 to 2021, increased significantly in 2022, and declined again in 2023**.
- The business experienced fluctuations rather than consistent year-over-year growth.

Sales performance appears to be influenced by changing market conditions, customer demand, or promotional activities. **The decline in 2023 may indicate slowing demand or reduced sales effectiveness.**

Top Province with The Highest Transactions Volume

Insight

- **West Java** contributes the highest transaction volume by a significant margin.
- Transaction volume in West Java is substantially higher than in other provinces.

West Java represents Kimia Farma's largest customer market and plays a critical role in overall business performance.

Analysis & Insights

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Top Provinces by Nett Sales

Insight:

- West Java also generates the highest net sales, followed by North Sumatra, Central Java, East Java, and North Sulawesi.
- The ranking is generally consistent with transaction volume, indicating that high sales are driven primarily by higher transaction activity.

These provinces contribute significantly to company revenue and should remain strategic business priorities.

Top Branches with Highest Branch Rating but Lowest Transaction Rating

Insight:

Several branches receive excellent overall branch ratings (4.8–5.0), while their average transaction ratings remain relatively low.

Customers appreciate the branch overall but are less satisfied with the transaction experience. This suggests operational issues rather than infrastructure or branch reputation.

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Analysis & Insights

Geographic Distribution of Nett Profit

Insight:

- West Java contributes the highest total profit among all provinces.
- Profit generation is concentrated in a limited number of provinces, while many regions contribute relatively lower profit.

Kimia Farma's profitability is geographically concentrated, increasing dependence on a few high-performing markets.

Conclusion

Key Findings

- Net sales fluctuated between 2020 and 2023, indicating inconsistent business growth.
- West Java is the largest contributor to transaction volume, net sales, and overall profitability.
- Customer satisfaction with branches (4.4) exceeds satisfaction with transaction experiences (4.0), highlighting opportunities to improve operational service quality.
- Business performance remains concentrated in several major provinces, suggesting potential for market expansion in lower-performing regions.

Strategic Recommendation

Business Recommendations

- Improve transaction efficiency and customer service quality across branches.
- Strengthen marketing investments and inventory planning in high-performing provinces such as West Java to maximize revenue potential
- Investigate the factors contributing to declining sales performance in 2023 and implement targeted promotional strategies to stimulate demand.
- Develop targeted improvement programs for underperforming regions by analyzing local customer demand and operational performance.
- Leverage historical business data for sales forecasting, demand planning, and strategic resource allocation to support long-term business growth.

Link Project & LinkedIn

https://github.com/grace-nc/kimiafarma-business-performance-analysis

https://datastudio.google.com/reporting/162035e9-b789-43eb-8c4e-bb0dde04e1a7

https://www.linkedin.com/in/grace-natalie-catherine

Thank You

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